

Subject Description Form

Subject Code	ITC3050I
Subject Title	Intimate Apparel Production Engineering
Credit Value	3 credits
Level	3
Pre-requisite/ <Co-requisite> / (Exclusion)	Nil
Objectives	This subject aims to provide students with technical knowledge of intimate apparel production processes, including cutting, sewing, welding, etc. and their respective technical problems during manufacturing. It equips students with analytical and critical thinking skills to evaluate and enhance the production processes for optimal efficiency and cost-effectiveness. This subject also helps the students to build first-hand experience of the latest technologies and real- life industrial practice so as to respond to the changing demands of the industry.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Manage, exploit and evaluate the production techniques and processes for intimate apparel, including spreading, cutting, sewing, welding, etc. - Demonstrate an integrated and critical approach to the processes and technologies involved in intimate apparel production so as to solve problems and achieve the required quality standards within an ever changing international and local market environment. - Proactively and critically evaluate the latest technologies and new developments from an industrial engineer perspective and within the context of global developments and trends.
Subject Synopsis/ Indicative Syllabus	(I) Cutting Principal processes in the cutting room, marker planning, spreading, cutting, sorting, material utilization in relation to intimate apparel production. Evaluation of material usage / wastage calculation. Application of cutting technology and integration of computer systems at cutting room. Analysis of the quality aspects of cutting room processes. (II) Sewing Sequence of assembly and flow of operations for various types of intimate apparel products. Appreciation of the characteristics of various apparel production systems. Application of specific sewing machinery and attachments for intimate apparel production. Quality aspects and specification of sewing operations. (III) Welding assembling method

	Principles, methods, technical aspects and control of welding for intimate apparel products. Quality aspects and inspection of weld seams and products. (IV) Industrial Engineering Major production concepts of workflow, balancing, throughput, cycle time, work-in-process. Line balancing techniques used in production: strengths and limitations. Lean production as a new manufacturing strategy for Fast Fashion. Principal and application of Standard Minute Value (SMV) in cost and production planning. Optimization of production scheduling and capacity planning. Fundamentals of method study, workplace engineering and time measurement. Techniques and systems for elimination of material handling. Applications of engineering, psychological, and physiological principles to the analysis and design of human work systems.																																						
Teaching/Learning Methodology	Lectures are designed to provide the general framework and principles of various intimate apparel production technologies. Professional seminars will help students to build their first-hand knowledge and related technical problems in specific production processes. Tutorials will be used to supplement formal lectures and seminars for further exploration of concepts and principles. Factory visit will be employed to enhance students' understanding of the latest technologies and real-life industrial practice.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th></th><th></th><th></th></tr><tr><td>1. Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assessment instrument will include a group project that students' ability to exploit and integrate the processes and technologies, tackle quality problems, as well as evaluate new developments in the market will be assessed. The knowledge building sheet and in-class discussion during tutorials aim to provide students a better understanding of work-based environment and real-life problem scenarios in the production processes. Students' performance in terms of problem-solving techniques, critical thinking and analytical skills to be applied in the industry will also be assessed by written examination.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c				1. Continuous Assessment	50%	✓	✓	✓				2. Examination	50%	✓	✓	✓				Total	100%						
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1. Continuous Assessment	50%	✓	✓	✓																																			
2. Examination	50%	✓	✓	✓																																			
Total	100%																																						

Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial/ Studio	12 Hrs.
	Other student study effort:	
	▪ Assignments	66 Hrs.
	Total student study effort	104 Hrs.
Reading List and References	<u>Book</u> Carr, H and Latham B. (2000). <i>The Technology of Clothing Manufacture</i>, Oxford, Blackwell Scientific Publications. Crowson R. (2006). <i>Factory Operations: Planning and Instructional Methods</i>. CRC Press. Hunter, A., King R. and Lowson, R.H. (2002). <i>The Textile/Clothing Pipeline and Quick Response Management</i>. Manchester [England]: Textile Institute, c2002 Meyers, F.E. and Stewart, J.R. (2002). <i>Motion and Time Study for Lean Manufacturing</i>. Upper Saddle River, N.J.: Prentice Hall, 3rd Edition. Yu, W., Fan, S.C., Harlock, S.C. & Ng, S.P. (2006). <i>Innovation and Technology of Women's Intimate Apparel</i>, The Textile Institute, Woodhead. <u>Supplementary</u> Glock, Ruth E. (2005). <i>Apparel Manufacturing: Sewn Product Analysis</i>, Upper Saddle River, N.J.: Prentice Hall, 4th Edition. Lausch, G. & Chazin, M. (2003). <i>Move Up to Improved Production, Upholstery Design and Management</i>, 16(10): 54-56. Salvendy, G. (2001). <i>Handbook of Industrial Engineering: Technology and Operations Management</i>. New York, NY: Wiley, 3rd Edition. Solinger, J. (1988). <i>Apparel Manufacturing Handbook</i>. Textile Book Publication. Swedberg, J. (2002). <i>So It Seems, Industrial Fabric Reports Review</i>. 79 (11): 46-49. <i>Welding Equipment: A Quiet Components of Success</i>, Geotechnical Fabrics Report, 2004, 22(6):41-42. <u>Websites</u> http://www.sonobondultrasonics.com http://www.gerbertechnology.com http://www.lectra.com http://www.eastmancuts.com http://www.shimaseiki.co.jp	